ASSIGNMENT 2 – CO2

Student details

Item	Details
Name	M. Manoj Kumar
Register number	192425114
Course	ITA03 – Mobile Computing
Assignment	Assignment 2 – CO2 (Multiple access and mobile communication)

Note on method: all simulation results in Q1, Q2 (Dataset B), Q3, Q4, Q5 and Q7 were produced by Python/NumPy programs with the random seed fixed to my register number 192425114, so every figure can be reproduced. Overheads, traffic models, SNR values and ratings are stated modelling assumptions, not quoted standard parameters.

Q1 – Mobile Communication or Mobile Computing?

1.1 Classification criteria

Criterion	Test applied
User mobility	Can the user or device move while the service stays active?
Access medium	Is the link wired or wireless?
Disconnection tolerance	Can the application keep working and recover when there is no connectivity?
Context awareness	Does the application use location, time or surroundings?
Resource constraints	Are battery, CPU, memory or radio resources limited?

1.2 Classification of the five cases

Case	Network type	Main problem area	Justification
a) Rugged tablet fetching work orders over cellular	Mobile, wireless	Both; mostly mobile computing	The worker moves between sites and cellular gives the link, but the app must handle session continuity, battery, security and changing signal quality.
b) Laptop on Ethernet at different offices	Nomadic, wired	Computing, not mobile communication	The laptop is moved between visits but is fixed and wired whenever it is in use, so there is no wireless mobility problem to solve.
c) Fixed wireless router at a remote substation	Fixed wireless	Mobile communication (wireless access)	The endpoint never moves; only the access link is radio. It is wireless but not mobile.
d) Push-to-talk handset	Mobile, wireless	Mobile communication	Moving users need low-latency, half-duplex voice over a shared radio channel with fast call set-up.
e) Tablet logs meter readings offline and syncs later	Mobile, intermittently connected	Mobile computing	The difficulty is local storage, queued updates, conflict handling and eventual synchronisation, not the radio link itself.

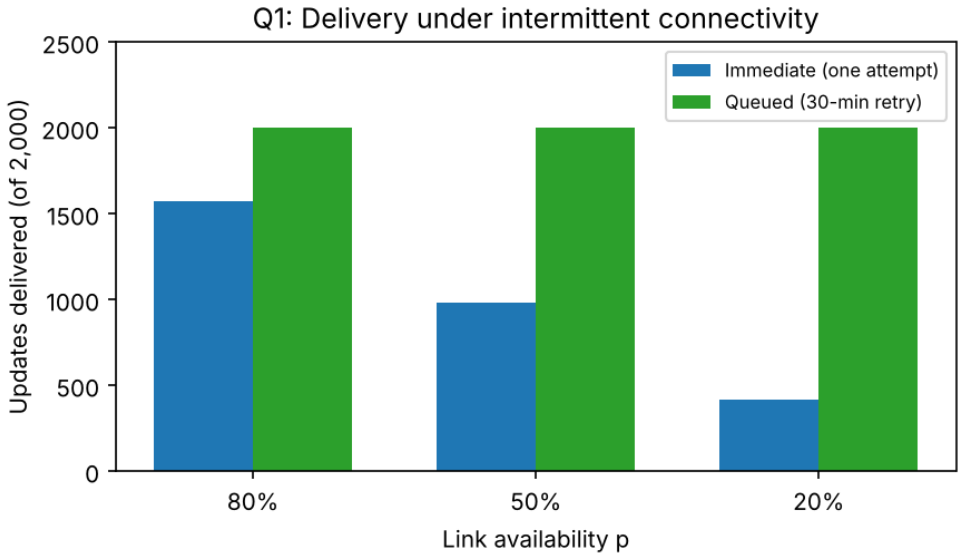
1.3 One additional ambiguous case

A vehicle-mounted GPS data-logger records a delivery truck's route while it passes through tunnels and dead zones and uploads the whole trip only when the truck returns to the depot Wi-Fi. The device is mobile and context-aware (location and time), but it uses no connection while moving. It is therefore a delay-tolerant mobile-computing system with only occasional wireless access. This shows that mobility, wireless access and mobile computing are three separate dimensions.

1.4 Store-and-forward simulation for case (e)

A workload of 2,000 meter-reading updates was simulated for link availabilities p = 80%, 50% and 20%. In the immediate strategy each update is sent once and lost if the link is down. In the queued strategy a failed update stays in the tablet's queue and is retried once per minute for up to 30 minutes. Each attempt succeeds independently with probability p.

Link availability	Immediate delivered	Immediate lost	Queued delivered (30 min)	Queued lost	Mean queued delay (min)	Theory (1−p)/p
80%	1572	428	2000	0	0.27	0.25
50%	979	1021	2000	0	1.05	1
20%	419	1581	2000	0	4.04	4



Interpretation: immediate sending delivers about p × 2,000 updates (1,572, 979 and 419 against expected 1,600, 1,000 and 400). Queuing delivered all 2,000 updates in every run because the probability of failing 31 straight attempts is negligible even at p = 20% (0.8^31 ≈ 0.001, i.e. about 2 updates expected for 2,000, and none occurred in this seed). The price is delay: the measured means of 0.27, 1.05 and 4.04 minutes match the theoretical expected waiting time (1−p)/p.

1.5 Conclusion

A wireless link alone does not give continuity, reliable delivery or consistent data. Cases (a) and (e) need application-level support: offline storage, retry queues, acknowledgements, duplicate suppression, conflict resolution and secure synchronisation. Mobile communication supplies connectivity; mobile computing adapts the application and its data workflow to mobility and disconnection.

Q2 – Sharing the Channel: FDMA vs TDMA Under Different Traffic

2.1 Nominal and usable capacity

FDMA: 2 MHz ÷ 25 kHz = 80 channels. TDMA: 2 MHz ÷ 200 kHz = 10 carriers, each with 8 slots, giving 10 × 8 = 80 slots. The nominal counts are equal, but the overheads are different, so the usable capacity is not.

Scheme	Nominal resources	Assumed overhead	Usable traffic resources
FDMA	80 channels of 25 kHz	4 channels for guard bands and control (5%)	76 channels
TDMA	10 carriers × 8 slots = 80 slots	1 slot per carrier per frame for synchronisation and control (10 of 80)	70 slots

FDMA wastes spectrum continuously in guard bands, while TDMA wastes time in guard periods and synchronisation. These percentages are my assumptions; real values depend on the air interface.

2.2 Traffic-condition predictions

Traffic	FDMA behaviour	TDMA behaviour	Predicted better
Continuous voice	One channel is held for the whole call; simple and predictable.	One slot per frame is held for the call; predictable delay.	FDMA, narrowly (76 vs 70 usable)
Bursty data, 10% duty	Dedicated channels sit idle 90% of the time.	Slots can be given only to devices that have data.	TDMA with dynamic slot assignment
70% voice / 30% data	Voice fills channels; data bursts leave reserved capacity idle.	Voice gets periodic slots, data uses leftover slots.	TDMA with separate QoS queues

2.3 Three designed traffic datasets

Dataset	Traffic model	Predicted / calculated result
A – Continuous voice	80 callers each need one resource for the full period.	FDMA serves 76 and blocks 4. TDMA serves 70 and blocks 10. Utilisation is 100% of usable resources in both.
B – Bursty telemetry	200 devices, each active in 10% of frames, 70 pooled slots per frame (100,000-frame simulation).	Dedicated allocation blocks 124 devices (FDMA) or 130 devices (TDMA) permanently. Pooled TDMA: mean active devices per frame 20.0 (s.d. 4.3, maximum 41), so demand never exceeded 70 slots in the simulation and nobody was blocked.
C – Edge case: synchronised burst	All 200 devices send at the same instant once a minute.	Dedicated schemes block the excess users. A pooled scheduler with a queue serves 70 per frame, so all 200 are delivered in ⌈200/70⌉ = 3 frames.

For Dataset B, the 76 FDMA channels carrying devices that transmit 10% of the time are busy only 7.6 channel-equivalents on average, so about 90% of the reserved capacity is idle. The pooled scheduler needs only about 20 slots on average.

2.4 Telemetry design decision

For a campus telemetry system where 90% of devices send a 200-byte report per minute, dedicated FDMA or TDMA allocation is not justified. The traffic is light and bursty, so a shared, scheduled channel with short transmission opportunities, acknowledgements and a queue is more efficient. Dedicated reservation is justified only if strict deterministic latency or guaranteed isolation is required.

2.5 Conclusion

With equal nominal resources, FDMA and TDMA differ in overhead and multiplexing flexibility. For continuous voice they are close (FDMA slightly ahead under my overhead assumptions). For bursty telemetry, dynamic TDMA is clearly better because idle time is shared.

Q3 – CDMA: Orthogonality, Spreading and the Near–Far Problem

3.1 DS-CDMA operation

In direct-sequence CDMA each BPSK bit $b \in \{-1,+1\}$ is multiplied by a user-specific code $c[k] \in \{-1,+1\}$ of L chips, so the transmitted chips are $b \cdot c[k]$. The receiver multiplies the received chips by the wanted user's code and sums over the L chips (despreading); a positive sum decides +1 and a negative sum decides −1. Processing gain is $G_p = L$, or $10 \cdot \log_{10}(L)$ dB.

Code length L	Processing gain (linear)	Processing gain (dB)
8	8	9.03 dB
16	16	12.04 dB
64	64	18.06 dB

3.2 Predictions before simulation

Condition	Prediction	Reason
Synchronous Walsh codes, equal power	BER stays at the noise-only level for all loads up to L−1 users.	Distinct aligned Walsh codes have zero cross-correlation.
Random codes, asynchronous users	BER rises steadily with the number of users.	Non-zero cross-correlation causes multiple-access interference (MAI); its variance is about 1/L per interferer.
One user at 10× power	The weak user's BER is much worse than with equal power.	The strong signal's residual cross-correlation is amplified by 10 in power; this is the near–far problem.

3.3 Correlation and receiver decision

The normalised correlation is $\rho_{ij} = (1/L) \Sigma c_i[k]c_j[k]$. Synchronous Walsh codes give $\rho_{ij} = 0$ for $i \neq j$; random binary codes give a mean of 0 and a variance near 1/L, so longer codes reduce typical interference. The matched-filter statistic for user 1 is $z_1 = (1/L) \Sigma r[k]c_1[k]$, which contains the wanted bit, the interference $\Sigma A_j b_j \rho_{j1}$ and noise. With chip offsets between users, Walsh codes lose their zero cross-correlation.

3.4 Simulation design and results

Method: 4,000 random BPSK bits per user, spreading with Walsh or random ±1 codes, amplitudes set per user, random chip offsets for the interfering users in the asynchronous cases, additive Gaussian noise at Eb/N0 = 7 dB for the reference user (noise-only BER ≈ 8×10^{-4}), and BER counted for the reference user after despreading. L = 8, 16, 64 were run with the same seed. In the near-far case user 2 transmits with 10× the power of the others.

Code length L = 8 (Gp = 9.03 dB)

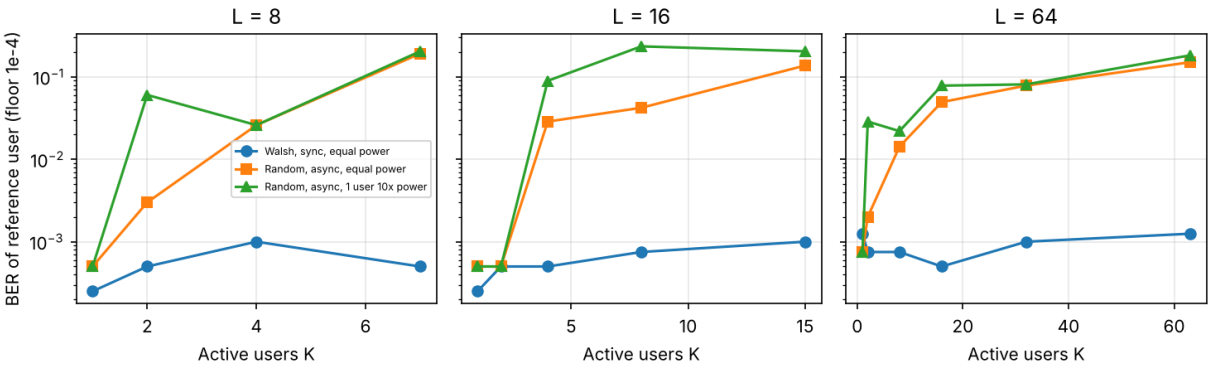
Active users K	Walsh, synchronous, equal power	Random, asynchronous, equal power	Random, async, one user 10× power
1	2.5e-4	5.0e-4	5.0e-4
2	5.0e-4	3.0e-3	6.0e-2
4	1.0e-3	2.6e-2	2.6e-2
7	5.0e-4	1.9e-1	2.0e-1

Code length L = 16 (Gp = 12.04 dB)

Active users K	Walsh, synchronous, equal power	Random, asynchronous, equal power	Random, async, one user 10× power
1	2.5e-4	5.0e-4	5.0e-4
2	5.0e-4	5.0e-4	5.0e-4
4	5.0e-4	2.9e-2	9.0e-2
8	7.5e-4	4.2e-2	2.4e-1
15	1.0e-3	1.4e-1	2.0e-1

Code length L = 64 (Gp = 18.06 dB)

Active users K	Walsh, synchronous, equal power	Random, asynchronous, equal power	Random, async, one user 10× power
1	1.3e-3	7.5e-4	7.5e-4
2	7.5e-4	2.0e-3	2.9e-2
8	7.5e-4	1.4e-2	2.2e-2
16	5.0e-4	5.0e-2	7.9e-2
32	1.0e-3	7.9e-2	8.1e-2
63	1.3e-3	1.5e-1	1.8e-1



Findings: (1) With synchronous Walsh codes the BER stays at about 10^{-3} (the noise floor) even at K = L−1, which confirms orthogonality. (2) With random asynchronous codes the BER grows with K: for L = 8, from 5×10^{-4} at K = 1 to 0.19 at K = 7; for L = 64, from 8×10^{-4} to 0.15 at K = 63. Longer codes tolerate more users at the same BER (for example BER ≈ 0.026 at K = 4 for L = 8, versus 0.014 at K = 8 for L = 64). (3) Adding a 10× user raises the weak user's BER sharply at small loads, for example at K = 2: 0.06 (L = 8) and 0.029 (L = 64) against 0.003 and 0.002 with equal power. The exact values depend on the particular random code pair drawn in this single-seed run, so they show a trend, not a precise law.

3.5 Simultaneous-user estimate and power control

For a target BER of 10^{-2} at Eb/N0 = 7 dB, my measured data show that random asynchronous codes support about 2 users for L = 8 and L = 16 (K = 4 gives 0.026 and 0.029 respectively) and somewhat fewer than 8 users for L = 64 (K = 8 gives 0.014). So for uncoordinated asynchronous uplinks the practical number of equal-power users is roughly L/8 or less, much lower than the L−1 possible with perfectly synchronised Walsh codes. This is a result of my simplified simulation, not a general CDMA capacity formula; real capacity also depends on voice activity, sectorisation, coding and fading.

Fast closed-loop power control is essential in the uplink: it tries to make all received powers equal at the base station, so a nearby handset cannot swamp a distant one even though their codes differ.

3.6 Conclusion

CDMA trades bandwidth for processing gain and code-domain separation. Orthogonality holds only with exact timing and channel assumptions. Asynchronism, multipath and unequal received powers create MAI, so synchronisation, RAKE multipath combining and accurate power control are central to reliable uplink operation.

Q4 – OFDMA: Giving Each User the Subcarriers They Can Use Best

4.1 Subcarrier spacing and cyclic prefix

For an OFDM symbol with useful duration Tu the subcarrier spacing is Δf = 1/Tu. For two subcarriers m ≠ n, (1/Tu)∫0^Tu exp(j2π(m−n)Δf t) dt = 0, and for m = n it equals 1, so the subcarriers are orthogonal over the useful interval. A cyclic prefix copies the tail of the symbol to its front. If the prefix is at least as long as the channel's delay spread, the channel acts as a circular convolution over the FFT window, which removes inter-symbol interference and keeps the subcarriers orthogonal.

4.2 Allocation approaches and predictions

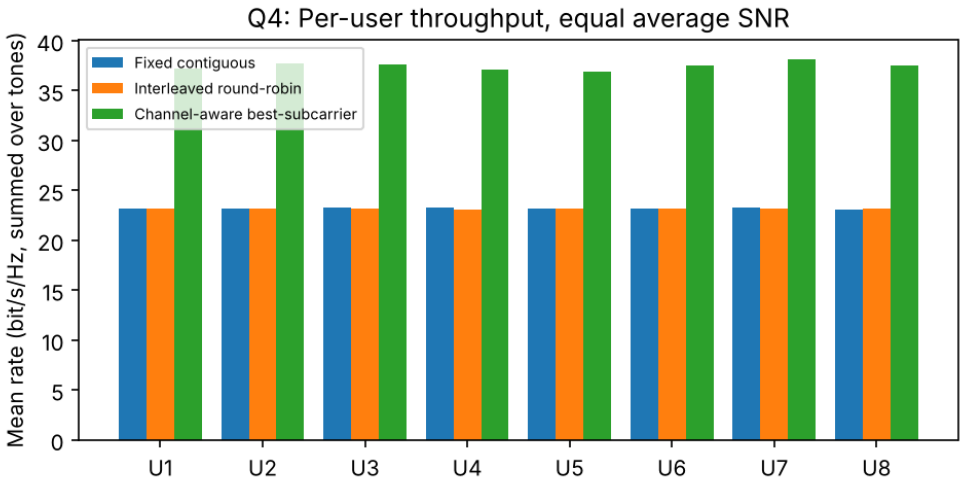
Approach	Rule	Prediction
Fixed contiguous	Each of 8 users gets 8 adjacent subcarriers.	Simple; a deep fade can hit one user's whole block. Fair in tone count, throughput may vary.
Interleaved round-robin	User u gets tones u, u+8, u+16, …	Frequency diversity; balanced fairness.
Channel-aware best-subcarrier	Each tone goes to the user with the highest gain on it.	Highest sum throughput; weak users may be starved.

4.3 Comparative experiment and measured results

1,000 independent realisations, 8 users, 64 subcarriers, Rayleigh power gains g ~ Exponential(mean 1), seed 192425114. Rate proxy r = log2(1 + 10g) bit/s/Hz (average SNR 10, linear). Jain's index J = (ΣRu)² / (8 ΣRu²) was computed in each realisation and averaged.

Scheme	Mean sum throughput	Lowest user mean	Mean Jain index
Fixed contiguous	185.64	23.03	0.9781
Interleaved round-robin	185.44	23.06	0.9780
Channel-aware best-subcarrier	299.74	36.90	0.9041

Scheme	U1	U2	U3	U4	U5	U6	U7	U8
Fixed contiguous	23.19	23.23	23.25	23.29	23.2	23.19	23.27	23.03
Interleaved round-robin	23.19	23.23	23.21	23.06	23.17	23.22	23.14	23.23
Channel-aware best-subcarrier	37.18	37.71	37.59	37.14	36.9	37.53	38.17	37.53



Findings: fixed contiguous and interleaved allocations give almost identical results (about 185.5 bit/s/Hz total, about 23.2 per user, Jain ≈ 0.978) because all users have the same average SNR. Channel-aware allocation raises the sum throughput by about 61% to 299.7. Its per-realisation fairness is lower (Jain 0.904), since the tones in any single realisation are shared unevenly. Over 1,000 realisations the users still average about 37 each, because here every user has the same average channel quality. This is the best case for opportunistic scheduling; Q7 shows what happens when users have unequal channels.

4.4 Lecture-hall allocation decision

I would use proportional-fair scheduling with a minimum-rate floor: first give enough subcarriers to any student whose average rate is below the floor, then give the remaining tones to the user with the highest instantaneous rate divided by past average rate. This keeps most of the opportunistic gain while guaranteeing every student a service level. Q7 tests this design.

4.5 Conclusion

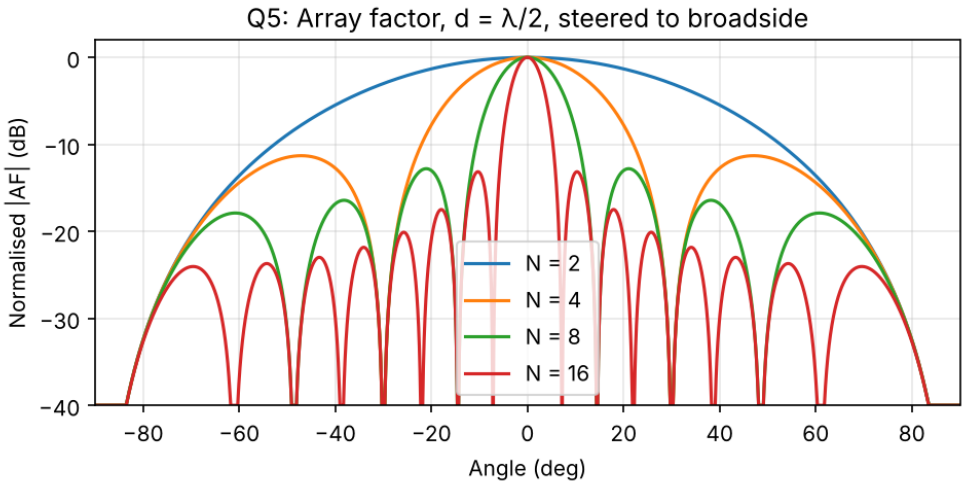
Channel-aware OFDMA maximises throughput and is fair only when users' channels are statistically equal. Round-robin or proportional-fair scheduling with a floor is more suitable when each student must get guaranteed service.

Q5 – SDMA: Reusing the Same Channel in Different Directions

5.1 ULA array factor and measured beam properties

For an N-element uniform linear array with spacing d, $AF(\theta) = \Sigma \exp[j\, n\, (kd \sin\theta + \beta)]$, with $\beta = -kd \sin\theta_0$ for steering to θ_0. The normalised magnitude is $|\sin(N\psi/2) / (N \sin(\psi/2))|$ with $\psi = kd(\sin\theta - \sin\theta_0)$. With $d = \lambda/2$ and steering at broadside, I evaluated the pattern numerically on a 0.005° grid.

N	Half-power beamwidth	First null	First sidelobe level
2	60°	none (≥ 90°)	no sidelobe
4	26.32°	30°	-11.3 dB
8	12.8°	14.48°	-12.8 dB
16	6.35°	7.18°	-13.15 dB



The beamwidth falls roughly as 1/N (60° → 26.3° → 12.8° → 6.35°). The sidelobe level approaches the ideal −13.26 dB of a uniform array as N grows; N = 2 has no sidelobes at d = λ/2.

5.2 SIR and angular separation

Setup: the beam is steered at user B (0°) and user A is at angular separation s, both with equal received power. The desired-to-interference ratio is SIR = 1/|AF(s)|², evaluated for s from 0.5° to 60° in 0.025° steps. Threshold: 15 dB.

N	First separation where SIR ≥ 15 dB	Separation beyond which SIR stays ≥ 15 dB
2	not reached within 60°	not reached within 60°
4	25.3° (at a pattern null)	not reached within 60°
8	12.2° (at a pattern null)	24.6°
16	6.1° (at a pattern null)	11.9°

A pattern null gives a deep but narrow region of isolation, so it is not enough for moving users. The reliable separation is the second column: about 24.6° for N = 8 and 11.9° for N = 16. For N = 2 and N = 4 the 15 dB target is not met reliably within 60°, because the sidelobes of 4 elements are only about 11 dB down.

5.3 Practical limits

SDMA is limited by angular spread from multipath, calibration errors, mutual coupling between elements, high sidelobes and users who share almost the same direction. If two users have nearly the same angle of arrival, one array cannot form two distinct spatial signatures for them.

5.4 Toll-plaza decision

At a toll plaza vehicles approach from nearly the same direction, usually within a few degrees. From Section 5.2, even 16 elements need about 12° of separation, so SDMA cannot be the only access method. I would use sectorisation and time or frequency scheduling as the main separation, and add beamforming as a link-quality improvement, not a guarantee of simultaneous service.

5.5 Conclusion

More elements give a narrower main beam and allow closer user spacing, but sidelobes and angular spread remain. SDMA works best when users are well separated in angle and the channel is stable.

Q6 – Choosing Multiple-Access Techniques for Real Deployments

6.1 Weighted decision criteria

Criterion	Weight	Meaning
Spectral efficiency	25%	Useful bits or users per unit bandwidth
Interference robustness	20%	Tolerance to interference, fading and imperfect synchronisation
Latency / QoS	20%	Suitability for voice, real-time traffic or periodic reports
Scalability	20%	Ability to handle growing numbers of users
Complexity / energy	15%	Implementation effort and device power (5 = simple and efficient)

Ratings use a 1–5 scale (5 = most suitable). Weighted score = Σ(weight × rating) / 100. The ratings are my engineering judgements based on Q2–Q5, not universal facts.

6.2 Ratings and weighted scores by deployment

Wide-area cellular voice

Technique	Spectral eff. (25%)	Robustness (20%)	Latency/QoS (20%)	Scalability (20%)	Complexity/energy (15%)	Weighted score
SDMA	4	3	4	4	2	3.50
TDMA	4	4	5	3	4	4.00
FDMA	2	3	4	2	4	2.90
CDMA	4	5	4	4	3	4.05
OFDMA	5	4	4	5	3	4.30

Top selection: OFDMA. OFDMA with spatial beamforming and QoS scheduling for voice.

Jam-resistant tactical link

Technique	Spectral eff. (25%)	Robustness (20%)	Latency/QoS (20%)	Scalability (20%)	Complexity/energy (15%)	Weighted score
SDMA	3	2	3	2	2	2.45
TDMA	2	3	4	3	4	3.10
FDMA	2	2	3	2	4	2.50
CDMA	4	5	4	3	2	3.70
OFDMA	3	3	3	3	3	3.00

Top selection: CDMA. CDMA / spread spectrum as primary, with frequency agility where permitted.

High-rate stadium downlink

Technique	Spectral eff. (25%)	Robustness (20%)	Latency/QoS (20%)	Scalability (20%)	Complexity/energy (15%)	Weighted score
SDMA	4	3	3	4	2	3.30
TDMA	3	3	3	3	4	3.15
FDMA	2	3	3	2	4	2.70
CDMA	3	4	3	4	2	3.25
OFDMA	5	4	4	5	3	4.30

Top selection: OFDMA. OFDMA with SDMA beamforming for spatial reuse.

5,000 low-rate sensors

Technique	Spectral eff. (25%)	Robustness (20%)	Latency/QoS (20%)	Scalability (20%)	Complexity/energy (15%)	Weighted score
SDMA	2	2	3	3	2	2.40
TDMA	4	4	4	4	4	4.00
FDMA	3	3	3	3	4	3.15
CDMA	3	4	3	4	2	3.25
OFDMA	3	3	3	4	3	3.20

Top selection: TDMA. Scheduled TDMA, or random access plus TDMA, with short reports and sleep cycles.

6.3 Why systems combine techniques

Each technique separates users in a different dimension: FDMA in frequency, TDMA in time, CDMA by code, OFDMA by orthogonal subcarriers, SDMA by direction. They solve different problems, so hybrids improve both capacity and QoS. For example, OFDMA subcarriers can be scheduled in time and then reused by directional beams.

6.4 Sensitivity check

I changed each of the five weights by +20% and −20% (renormalising the weights each time, 10 cases per deployment) and recomputed the winner. The top choice did not change in any case: OFDMA for cellular voice, CDMA for the tactical link, OFDMA for the stadium and TDMA for the sensors. The least comfortable margin is cellular voice, where CDMA (4.05) is close to TDMA (4.00) behind OFDMA (4.30). A different set of ratings could still change that ranking.

6.5 Conclusion

No single technique is best everywhere. The right choice depends on traffic shape, mobility, interference, latency, energy budget and user density, and hybrid designs are preferable when several constraints apply together.

Q7 – Diagnose, Redesign and Validate

Selected limitation: channel-aware best-subcarrier allocation (Q4) starves users with weak average channels.

A. Diagnose the bottleneck

- Best-subcarrier allocation gives every tone to the instantaneously strongest user. If one user is consistently weaker (for example far from the base station), that user almost never wins a tone.
- In Q4 all users had the same average SNR, which hid the problem. For this test the 8 users have average SNRs of 30, 20, 12, 8, 5, 3, 2 and 1.2 (linear), from nearest to farthest.
- The quantity that collapses is the number of tones won by weak users, which drives their rate to nearly zero while the sum throughput looks excellent.

B. Design objective

Keep as much of the opportunistic throughput gain as possible while guaranteeing every active user a mean rate of at least 9.96 bit/s/Hz (summed over the 64 tones). This floor equals 50% of the average per-user rate under round-robin in the same scenario. The design must not change the bandwidth or the number of tones.

C. Revised strategy

Use proportional-fair (PF) scheduling with a minimum-rate floor. Each tone goes to the user with the largest metric r(u,k) / R̄(u), where R̄(u) is the user's exponentially averaged past rate (β = 0.02). Users whose R̄ is below the floor have their metric multiplied by 10, so they are served first.

Pseudocode:

```
FOR each scheduling slot:
    FOR each tone k:
        metric[u] = rate(u,k) / avg_rate[u]
        IF avg_rate[u] < floor: metric[u] = 10 * metric[u]
        assign tone k to argmax_u metric[u]
    FOR each user u: avg_rate[u] = (1 - beta)*avg_rate[u] + beta*rate_this_slot[u]
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The redesign adds per-user rate tracking, so the base station needs a running average for each user and the channel gain feedback it already needs for best-subcarrier scheduling.

D. Prediction before implementation

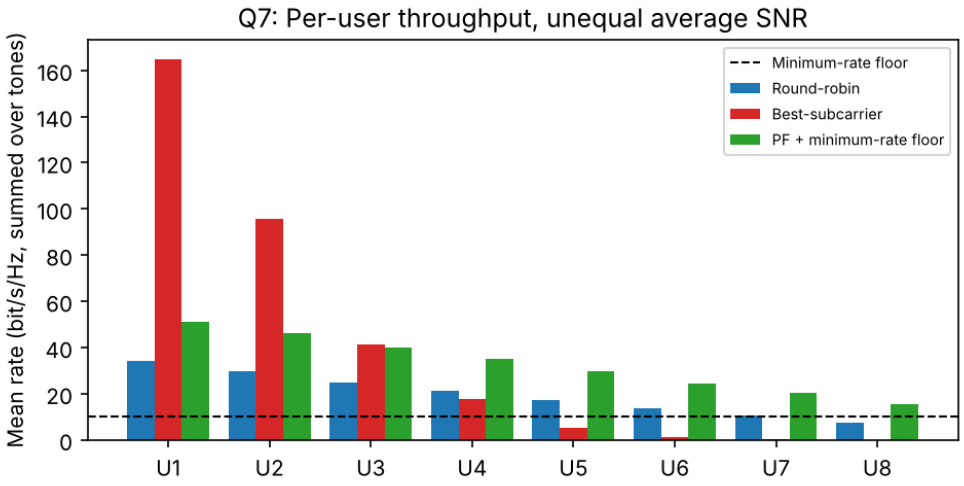
I expected that the weakest user's rate would rise from almost zero to above the floor, Jain's fairness over users' mean rates would rise strongly, and the sum throughput would fall below best-subcarrier but stay above round-robin.

E. Implementation and validation

1,000 slots, 8 users, 64 tones, seed 192425114 + 7, rate proxy log2(1 + SNR_u · g) with g ~ Exponential(1). Jain's index is computed over the users' mean rates.

Scheme	Sum throughput	Weakest user (U8) mean	Jain over mean user rates	All users ≥ floor (9.96)?
Round-robin	159.41	7.66	0.8392	No
Best-subcarrier	326.25	0.05	0.3467	No
PF + minimum-rate floor	262.73	15.65	0.8865	Yes

Scheme	U1	U2	U3	U4	U5	U6	U7	U8
Round-robin	34.06	29.83	24.94	21.33	17.34	13.63	10.6	7.66
Best-subcarrier	164.92	95.62	41.12	17.66	5.44	1.21	0.23	0.05
PF + minimum-rate floor	51.24	46.21	39.96	35.1	29.82	24.39	20.36	15.65



F. Defend the final design

The result matches the prediction. Best-subcarrier gives the highest sum (326.2 bit/s/Hz) but nearly eliminates the weakest user (0.05) and has Jain 0.347; the four weakest users are below the floor. PF with the floor lifts the weakest user to 15.65, so every user is above the 9.96 target, and Jain rises to 0.887. The sum throughput is 262.7: about 19.5% lower than best-subcarrier but about 65% higher than round-robin (159.4).

The trade-offs are the cost in total throughput compared with pure best-subcarrier and the extra state and tuning parameters (β and the boost factor of 10). Pure best-subcarrier remains better when only total throughput matters and all users have similar channels, as in Q4. These results come from a simplified Rayleigh model with fixed average SNRs; moving users and imperfect channel feedback should be tested before real deployment.

Overall conclusion

Mobile systems have to be designed around traffic behaviour and operating constraints, not just the wireless medium. Offline queues solve intermittent connectivity (Q1); pooled TDMA fits bursty traffic better than dedicated channels (Q2); CDMA depends on synchronisation and power control (Q3); OFDMA scheduling trades throughput against fairness, and a minimum-rate floor protects weak users (Q4, Q7); SDMA reuses the channel only where users are angularly separable (Q5); and a weighted comparison shows the techniques are complementary (Q6).

References

- ITA03 – Mobile Computing, Assignment 2 – CO2 question paper.
- S. Haykin, Communication Systems, 5th ed., Wiley – digital modulation and spread-spectrum concepts.
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Note: overhead percentages, traffic models, SNR values, rate floors and ratings are modelling assumptions chosen for this report, not quoted standard parameters.